

NATIONAL TELE-OPHTHALMOLOGY AI SCREENING PROGRAM

Explainable AI Clinical Decision Support System | Rural PHC Triage (SIH PS 26038)

Patient ID: IND-PHC-2026-0814 Age/Sex: 56 / Male Screening Date: 05-Sep-2026 08:32
PHC Health Centre: PHC Kolar District #04 Frontline Operator: ASHA Health Worker #108

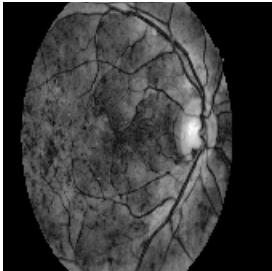
TRIAGE ACTION: URGENT SPECIALIST REFERRAL REQUIRED (<48 HOURS)

High risk of irreversible vision loss. Tertiary Vitreoretinal evaluation mandatory.

1. Raw Fundus Scan



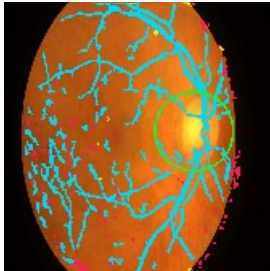
2. Enhanced Green



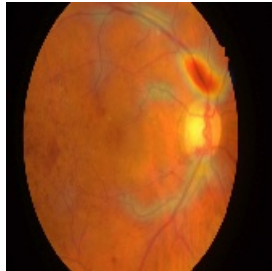
3. Retinal Vessels



4. Lesion Detections



5. Grad-CAM XAI

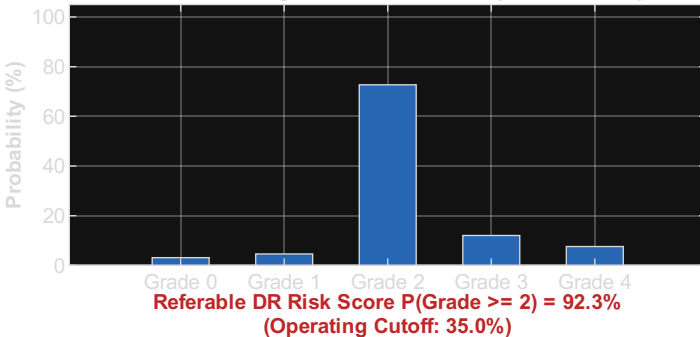


Legend: [Cyan]: Vascular Tree | [Yellow]: Hard Exudates (Lipids) | [Magenta]: Microaneurysms/Hemorrhages | [Jet Colormap]: AI Focus Regions

QUANTITATIVE BIOMARKER ANALYSIS

Image Quality Assessment (IQA): PASS (BRISQUE: 35.2)
Mean Illumination Level: 45.7% (Dynamic Range: 0.26)
Vascular Density (Coverage): 7.56% of FOV
Hard Exudates (Bright Lesions): 8 clusters (122 px area)
Microaneurysms (Early NPDR): 123 detected foci
Retinal Blot Hemorrhages: 4 detected sites
Optic Disc Status: Localized (Center: [361, 245])

AI DR Severity: Moderate NPDR (Conf: 72.7%)



OPHTHALMOLOGIST TELEMEDICINE REVIEW & VALIDATION (Time required: under 30 seconds)

Clinical Impression: ☐ Concordant with AI diagnosis ☐ Upgrade Severity ☐ Downgrade / Artefact
Action Prescribed: ☐ Tele-Consultation Booked ☐ Anti-VEGF / Laser Referral ☐ PHC Annual Rescreen
Reviewing Ophthalmologist Signature: _____ Date/Time: _____